

Image appears too bright:

Suitable method: Logarithmic transformation

Concept: Compression of high intensity pixel values and enhances darker regions.

Formula: $S = c \log(1+r)$

Process:

- 1) Read the image.
- 2) Compare pixel & apply log transformation.
- 3) Normalize the output intensity.

Outcomes:

→ Reduces excessive brightness & Improves visibility in darker regions.

2. Noisy image with random intensity Variations:

Suitable filter: Median filter

Concept: Replaces each pixel with the median value of its neighborhood.

Working:

- 1) select a 3×3 or 5×5 window
- 2) sort pixel values.
- 3) Replace the center pixel with the median.

Impact:

- Removes salt & pepper noise.
- Preserves edges better than averaging filters
- Produces a smoother image.

3. Blurred image after smoothing:

Suitable method: Unsharp Masking

Working:

- 1) Blur the image.
- 2) Subtract blurred image from original to obtain mask
- 3) Add the mask back to the original.

Formula:

$$g = f + k(f - f_{\text{blur}})$$

Expected Results:

- Restores sharp edges
- Improves fine details
- Enhances image clarity.

4. Fine details are not visible:

Suitable technique: High pass filter

Concept: Enhances high-frequency components such as edges & textures.

Working:

- 1) Apply a high-pass convolution kernel.
- 2) Suppress low frequency information.
- 3) Highlight sharp intensity changes.

Effect:

- Improves image sharpness
- Makes fine details more visible
- Enhances edges.

image Preprocessing before analysis

Suitable method: Gaussian filter

Concept: Smooths the image by reducing Gaussian noise.

Working:

- 1) Apply gaussian kernel convolution.
- 2) Replace each pixel with weighted neighboring values.

Justification:

- 1) Removes noise.
- 2) Preserve overall image structure.
- 3) Improves performance of later segmentation and edge detection.

6) Object - brighter than background

Suitable segmentation: Global thresholding.

Concept: Separates object & background using one threshold value.

Working:

- 1) Choose Threshold T .
- 2) Pixels $> T$ become foreground
- 3) Pixels $= T$ become background

Effectiveness:

- Simple & fast
- Works well when contrast is high.
- Produces a binary image.

7) Separate foreground & background on intensity
Suitable method: Otsu's Thresholding

Concept: Automatically selects the optimal threshold by minimizing intra-class variance.

Process:

- 1) Analyze image histogram
- 2) Compute optimal threshold
- 3) Segment into foreground & background.

Advantages:

- 1) Automatic
- 2) Accurate for bimodal histograms
- 3) No manual threshold selection.

8. Identify object edges:

Suitable method: Canny Edge detection

Working:

- 1) Gaussian smoothing
- 2) Gradient calculation
- 3) Non-maximum suppression
- 4) Double thresholding
- 5) Edge tracking by hysteresis.

Effectiveness:

- Detects accurate edges
- Reduces noise.
- Produces thin & continuous boundaries.

Detected edges are noisy and fragmented

Suitable method: Morphological operations

Concept: Refines detected edges using dilation followed by erosion.

Working:

- Dilation connects broken edges.
- Erosion restores edge thickness

Role in Segmentation:

- Removes small gaps.
- Produces continuous boundaries.
- Improves object detection accuracy.

Precise object boundary extraction

Suitable method: Canny Edge Detection + Active Contour

Working:

- Detect initial edges using canny.
- Active contour iteratively moves toward the true object boundary by minimizing energy.

Effectiveness:

- Accurate boundaries extraction.
- Handles complex object shapes.
- Suitable for medical imaging, industrial inspection, & object recognition.